

Title: The effectiveness of high intensity exercise therapy for improving bone density in postmenopausal women: a systematic review and meta-analysis protocol

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1. Introduction

With the increase in life expectancy throughout the world, populations everywhere are aging. Osteoporosis is a disease that presents with bone fragility which progresses with age (1). As bone mineral density (BMD) decreases due to osteoporosis, the risk of fractures due to falls increases (2). The incidence of osteoporosis is particularly high in women, due to the decrease in female hormones after menopause. Treatments for osteoporosis, lifestyle modification, dietary therapy, and pharmacological interventions are known to be effective (3). Exercise therapy has also been shown to be useful for the prevention and treatment of osteoporosis in women (4). A recent systematic review reported on changes in BMD with different exercise intensity (5). According to this review, the higher the intensity of exercise, the greater the improvement in BMD. However, subsequent to the publication of the review, there has been a growing trend of randomized controlled trials of higher intensity exercise therapy interventions that have not previously been implemented to avoid the risk of fracture (6)(7). The purpose of the current review is to examine the effects of high intensity exercise on BMD changes compared to moderate intensity exercise, and how the safety of exercise differs.

2. Research question

To determine the difference in effect on BMD changes at the lumbar spine and femoral neck between high intensity exercise therapy and moderate intensity exercise therapy in postmenopausal women.

P: postmenopausal women

I: high intensity exercise therapy

C: moderate intensity exercise therapy

O: BMD changes at the lumbar spine and femoral neck

3. Method

3.1 Protocol

In preparing this protocol, we adhered to the preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 (8). We aim to publish this protocol in protocols.io (<https://www.protocols.io/>).

3.2 Inclusion criteria for articles in the review

3.2.1 Study type

We will cover randomized controlled trials assessing the effectiveness of high intensity exercise therapy for improving bone density in postmenopausal women. There will be no country or language restrictions. All papers will be included, whether published or unpublished, as well as abstracts of conferences and letters. No animal studies will be included. We will not exclude any studies based on either the observation period or the year of publication.

3.2.2 Study participants

We will include studies that meet the criteria as follows: studies of interventions with postmenopausal women; participants with an osteoporotic condition or osteoporosis (T-score less than 1 standard deviation or less than 80% of young adult mean); participants in each group were treated by drug therapy for osteoporosis or were not treated with any drugs for osteoporosis; high intensity exercise therapy or moderate intensity exercise therapy was prescribed in any of the intervention groups; the outcome included the BMD of the lumbar spine and femoral neck.

We will exclude studies that meet the criteria for participants as follows: any diseases affecting bone metabolism; uncontrolled cardiovascular disease; cognitive disorder (Mini-Mental State Examination score less than 24); immediately following surgery or injury to the lower extremity (within a few months); immediately following an injury or localized pain in the lower back.

3.2.3 Intervention

High intensity exercise intervention in this review was defined when one of following criteria is met: 1 repetition maximum (1RM) of 80% or more, 6 times or less; ground reaction force (GRF) greater than or equal to 4 times body weight; bodyweight jumps with stiff-legged landing; aerobics performed with a load greater than 4 times body weight.

3.2.4 Control

Moderate intensity exercise intervention is allocated as control. It was defined when one of the following criteria is met: 1RM of 65-80%, 8-15 times; GRF greater than twice to less than 4 times body weight; heel drop exercise (dorsiflexion 0° or more, heel lift); aerobics performed with a load less than 4 times body weight.

If more than one exercise intervention is being used, the one with the highest intensity should be considered for allocation.

3.3 Types of outcomes

3.3.1 Primary outcomes

1. BMD of lumbar spine: DEXA (dual-energy X-ray absorptiometry)
2. BMD of femoral neck: DEXA (dual-energy X-ray absorptiometry)
3. Adverse events: falls, fractures, delayed onset myalgia, etc.
4. Quality of life: SF-36, etc.

3.3.2 Secondary outcomes

1. Lean body mass

3.4 Search method

3.4.1 Electronic search

The following databases will be searched:

1. The Cochrane Central Register of Controlled Trials (CENTRAL);
 2. PubMed;
 3. Cumulative Index to Nursing and Allied Health Literature (CINAHL);
 4. Web of Science;
 5. EMBASE via ProQuest Dialog;
 6. MEDLINE via ProQuest Dialog;
- (Refer Appendixes 1, 2, 3, 4, 5 and 6 for search strategies.)

3.4.2 Other resources

In addition, we will search these databases for unpublished or ongoing trials:

1. The World Health Organization International Clinical Trials Platform Search Portal (ICTRP);
2. ClinicalTrials.gov;

We will request authors of original studies to provide unpublished or additional data.

3.5 Data collection and analysis

3.5.1 Selection of studies

Two independent reviewers (K.H. and T.D.) will screen titles and abstracts, then, based on the full texts, assess their eligibility. We will contact authors for relevant data that is missing. Divergence of views between K.H. and T.D. will be discussed and resolved, or, failing this, a third reviewer (T.K.) will be the arbiter.

3.5.2 Data extraction and management

K.H. and T.D. will independently carry out data extraction from the eligible studies by means of a standardized data collection form: a pre-checked form that uses 10 randomly selected studies. The form will contain information on general characteristics (study ID, extraction date, title, authors, and source of study if not published), study design (detailed type of randomized controlled trial, participants, and exclusion criteria used in the study), study population, interventions (frequency, intensity, type, and time of exercise in the intervention group, details of the control group, and follow-up period), outcomes (changes in BMD of the lumbar spine or femoral neck after the exercise intervention, adverse events) and details needed to assess risk of bias. Again, any differing views will be put to discussion, with T.K. as arbiter, if required.

3.6 Assessment of risk of bias in included studies

Reviewers K.H. and T.D. will evaluate the risk of bias independently using the Risk of Bias 2. Each study will be reviewed for five criteria and recorded as low risk of bias, unclear risk of bias, or high risk of bias (9). The two reviewers will discuss any disagreements, with a third reviewer (S.Y.) as arbiter, if necessary.

3.7 Measures of treatment effects

We will pool the mean differences and the 95% CIs for these continuous variables: BMD of lumbar spine, BMD of femoral neck, quality of life, and lean body mass. If the studies included have employed several different scales, we will pool the effect estimates using standard mean differences. We will summarize adverse events using the definition in the original article, and, if possible, we will carry out meta-analysis.

3.8 Unit of analysis issues

For the crossover trials, data from only the first period will be incorporated. In dealing with cluster-randomized controlled trials, for dichotomous data, we will employ the design effect and determine effective sample size and number of events by applying the intraclass correlation coefficient (ICC) for each unit and the average size of cluster, as outlined in the Cochrane Handbook (Chapter 16.3.5) (10).

In the case that the ICC has not been reported, instead, we will apply the ICC of a similar study. For continuous data, the sample size only will be reduced, with means and standard deviation staying the same (10).

3.9 Dealing with missing data

We will request from the original authors data that was not presented.

3.9.1 Missing outcomes

We will carry out an intention-to-treat (ITT) analysis, to the extent it is possible, for all dichotomous data.

For continuous data, we will not use the recommendation in the Cochrane Handbook to impute missing data (10). Instead, we will carry out meta-analysis with regard to the available data in the original study.

3.9.2 Missing statistics

If original studies report only standard error or p-value, the standard deviation will be calculated based on Altman's method (11).

If these values are unknown to us when we contact authors, we will calculate standard deviation by t-value and confidence interval, using the method in the Cochrane Handbook(10), or employing a validated method (12). These methods will be analyzed for validation by sensitivity analysis.

3.10 Assessment of heterogeneity

To evaluate statistical heterogeneity, we will visually inspect the forest plots and calculate the I^2 statistic (I^2 values of 0% to 40%: may not be important; 30% to 60%: may indicate moderate heterogeneity; 50% to 90%: may indicate substantial heterogeneity; 75% to 100%: may indicate considerable heterogeneity). Where substantial heterogeneity is present ($I^2 > 50\%$), we will assess the reason for such heterogeneity. Cochran's χ^2 test (Q test) will be run for the I^2 statistic, and a p-value of less than 0.10 will be considered to be statistically significant.

3.11 Assessment of reporting bias

We will access the clinical trial registry system (ClinicalTrials.gov and ICTRP) to perform an extensive literature search for trials that are unpublished. Potential publication bias will be assessed by visually inspecting the funnel plot. The Egger test will also be performed. We will not run the test if we find fewer than 10 trials or trials with a similar sample size. We will assess the potential publication bias by visual inspection of the funnel plot.

3.12 Meta-analysis

Meta-analysis will be carried out with Review Manager software (RevMan 5.4), using a random-effects model.

3.13 Subgroup analysis

To explain the influence effect modifiers can have on results, we will conduct the subgroup analyses of the primary outcomes, based on the factors below, when we have gathered sufficient data:

1. We will divide the participants into two groups according to the age of the subject, either 60 years old or more, or younger than 60 years old, to examine the difference in effect by age.
2. We will divide the interventions into two types, based on the type of exercise (strength training or aerobic exercise) to examine the difference in effect by type of exercise type.
3. If the sample size is big enough, we will also examine the difference in the effect of medication on bone mineral density.

3.14 Sensitivity analysis

We planned to perform a sensitivity analysis for primary outcomes to determine whether high risk of bias of some included studies affected the study results. We defined 'high risk' based on inadequate allocation concealment, random sequence generation, and studies with missing data of greater than 20% (13). We planned to carry out the sensitivity analyses as follows:

1. Exclusion of studies with high risk of bias
2. Exclusion of cluster-RCTs
3. Comparing 95% CIs or fixed-effect pooled estimates versus 95% CIs or random-effects pooled estimates

4. Summary of findings table

This table will be generated for the following outcome, as per the Cochrane Handbook (10). Grading will be used to evaluate the quality of evidence, employing the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach for the summary of findings table (14). The quality of the evidence will be decided by one researcher (K.H.) and then confirmed and finalized by another researcher (T.K.).

5. Conflict of interest

The authors declare no conflict of interest.

6. Support

None.

Appendix 1: the Cochrane Central Register of Controlled Trials(CENTRAL) ;search strategy

Participant Keywords: (MeSH descriptor: [Menopause] explode all trees) AND (female):ti,ab,kw

Intervention Keywords: (MeSH descriptor: [Exercise] explode all trees) OR (MeSH descriptor: [Sports] explode all trees) OR (MeSH descriptor: [Physical Fitness] explode all trees) OR "physical activity"

Study design Keywords: ("randomized controlled trial" OR "controlled clinical trial" OR "randomised" OR "randomized" OR "placebo" OR "randomly" OR "trial")

Appendix 2: PubMed search strategy

Participant Keywords: ("female"[MeSH Terms] OR "female"[Title/Abstract]) AND ("menopause"[MeSH Terms] OR "menopause"[Title/Abstract])

Intervention Keywords: "exercise"[MeSH Terms] OR "exercise"[Title/Abstract] OR "exercise therapy"[MeSH Terms] OR "exercise therapy"[Title/Abstract] OR "sports"[MeSH Terms] OR "sports"[Title/Abstract] OR "physical fitness"[MeSH Terms] OR "physical fitness"[Title/Abstract] OR "physical activity"[Title/Abstract]

Study design Keywords: ("randomized controlled trial" [tiab]) OR ("controlled clinical trial"[tiab]) OR (randomised[tiab]) OR (randomized[tiab]) OR (placebo[tiab]) OR (randomly) OR (trial[ti]) OR "randomized controlled trial"[Title/Abstract] OR "controlled clinical trial"[Title/Abstract] OR "randomised"[Title/Abstract] OR "randomized"[Title/Abstract] OR "placebo"[Title/Abstract] OR "randomly"[All Fields] OR "trial"[Title]

Appendix 3: CINAHL search strategy

Participant Keywords: (MM "Menopause+") OR (MM "Postmenopause") OR TX postmenopaus* OR TX menopaus*

Intervention Keywords: (exercise OR sports OR "physical fitness" OR "physical activity")

Study design Keywords: ("randomized controlled trial" OR "controlled clinical trial" OR "randomised" OR "randomized" OR "placebo" OR "randomly" OR "trial")

Appendix 4: Web of Science search strategy

Participant Keywords: (female AND menopause)

Intervention Keywords: (exercise OR sports OR "physical fitness" OR "physical activity")

Study design Keywords: ("randomized controlled trial" OR "controlled clinical trial" OR "randomised" OR "randomized" OR "placebo" OR "randomly" OR "trial")

Appendix 5: EMBASE search strategy

Participant Keywords: (osteoporosis/ OR osteopor* OR osteopenia* OR bone density/ OR bone mass/ OR bone densit* OR bone loss* OR bone mass* OR bone mineral/ OR bone mineral densit* OR bone mineral content*) AND (postmenopause/ OR postmenopause osteoporosis/ OR menopause/ OR (menopaus* OR postmenopaus* OR post-menopaus* OR (post NEAR menopause*)))

Intervention Keywords: (exp exercise/ OR exp sport/ OR fitness/ OR exp physical activity/ OR sport* OR exercis* OR physical fitness* OR physical activit* OR vibration therapy/ OR vibration therap*)

Study design Keywords: (randomized controlled trial/ OR double blind procedure/ OR single blind procedure/ OR triple Blind procedure/ OR randomi*: tiab OR randomisation/ OR placebo/ OR placebo* OR ((controlled OR comparative OR placebo OR random*) NEAR/3 (trial OR study)) (random*NEAR/7 (allocate*OR allot*OR assign*OR basis*OR divid*or order*)) ((singl*OR doubl*OR trebl*OR tripl*) NEAR/3 (blind*OR mask*)))

Appendix 6: MEDLINE search strategy

Participant Keywords: (osteoporosis/ OR osteopor* OR osteopenia OR bone densit* OR bone OR bones OR bone loss* OR bone mass* OR bone mineral densit* OR bone mineral content* OR bone age OR bone defect* OR bone demineralization OR bone mineral* OR bone strength OR decalcifi* OR demineralized bone) AND (exp menopause/ OR (menopaus*OR postmenopaus*OR post-menopaus*OR (post adj menopause*)))

Intervention Keywords: (exp exercise/ OR exp exercise therapy/ OR exerci* OR exp sports/ OR sport* OR physical fitness OR physical activit* OR vibration/tu OR vibration therap*)

Study design Keywords: (randomized controlled trial OR controlled clinical trial OR randomi?ed:tiab OR placebo:ab OR clinical trials as topic OR randomly:tiab OR trial:ti NOT (animals not (humans and animals)))

Appendix 7: Reference

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